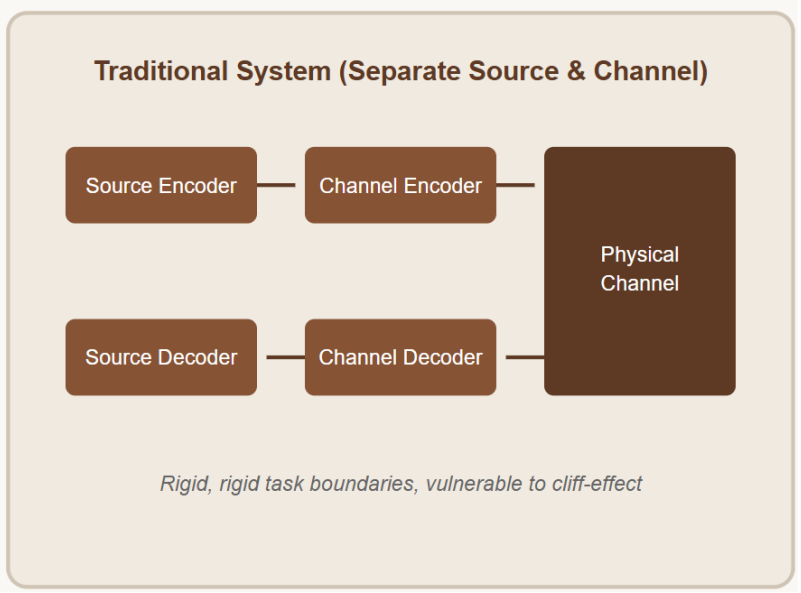
Task Aware Dynamic Semantic Compression Framework for 6G Deep JSCC

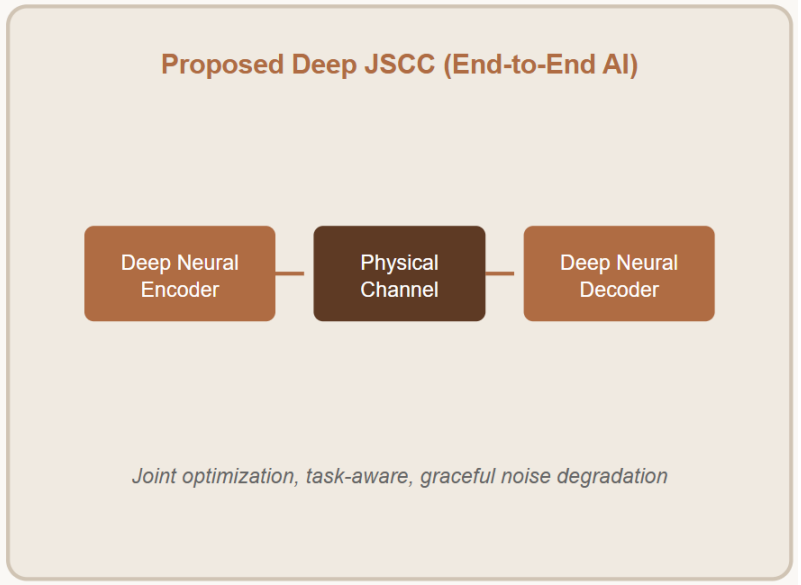
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Dr. Thanikaiselvan V.
Professor, SENSE

Introduction

- **6G shifts from bit-accurate delivery to meaning- and task-oriented delivery.** Raw camera and sensor data is too large for the radio link.
- **Cliff effect:** separate source and channel coding fails abruptly below a threshold SNR, while JSCC degrades gradually.
- **Receivers differ:** a human operator needs a viewable image; a machine (traffic AI) needs only task-relevant semantics.
- **Proposed system:** one transmitter, two modes.
- **Use case:** traffic imagery (highway vs urban).





Human mode: compressed latent, reconstructed image. **Machine mode:** 128-value semantic vector, classification.

Literature Review: Foundations, Surveys and Deep JSCC

Work	Contribution	Relevance to this project
Nguyen et al., IEEE COMST, 2025	Survey: theory of mind, generative AI and deep JSCC for semantic communication	Frames the shift to meaning-centric 6G
Huang et al., D²-JSCC, IEEE JSAC, 2025	Digital deep JSCC that works with practical digital transceivers	Bridges JSCC and real digital links
Yang et al., SwinJSCC, IEEE TCCN, 2025	Swin-Transformer JSCC, adaptive to SNR and rate	Candidate backbone for future work
Semantic-aided parallel image transmission, IEEE TWC, 2025	JSCC compatible with practical systems	Coexistence with conventional coding
Semantic base with fine-grained HARQ, IEEE TWC, 2025	Reliable semantic image transmission with HARQ	Reliability extension
Zia et al., Computer Networks, 2026	Survey on explainable AI for semantic communication	Explainability direction
Jain et al., IEEE Commun. Standards Mag., 2025	Standardization of AI-native 6G and semantic protocols	Places work in 6G standards context

Surveys and deep JSCC show gradual degradation and practical compatibility, but target a single objective.

Literature Review: Task-Oriented and Generative Approaches

Work	Contribution	Relevance to this project
Diao et al., IEEE JSAC, 2025	Aligns task- and reconstruction-oriented communications for edge intelligence	Closest to the dual-objective idea
DRL + information bottleneck, IEEE JSAC, 2025	Task-oriented semantic communication with IB and deep RL	Sends only task-relevant features
End-to-end learning over MIMO, IEEE JSAC, 2025	Information-theoretic task-oriented design over MIMO channels	MIMO extension
Semantics-guided diffusion for DeepJSCC, IEEE TWC, 2025	Diffusion guided by semantics improves wireless image quality	Upgrade for human mode
Semantic-prior latent diffusion, IEEE TWC, 2025	Channel-adaptive equalizing and denoising with latent diffusion	Latent-based, like our VAE branch
Visual-language-model cross-modal SC, IEEE TWC, 2025	Cross-modal semantic communication with a VLM	Richer semantic extraction
RSMA panoramic video semantic transmission, IEEE TWC, 2025	Adaptive semantic video transmission with RSMA	Multi-user extension

Most work optimizes reconstruction or a task, rarely both in one lightweight design with explicit bandwidth accounting.

Research Gap

- Existing JSCC work optimizes either pixel reconstruction (human) or task accuracy (machine), rarely both in one architecture.
- Many designs ignore the transmitted payload size, a major cost in 6G links.
- Digital pipelines have a threshold SNR below which they fail completely.
- No simple, reproducible, lightweight dual-mode demonstrator with explicit bandwidth accounting for traffic imagery.

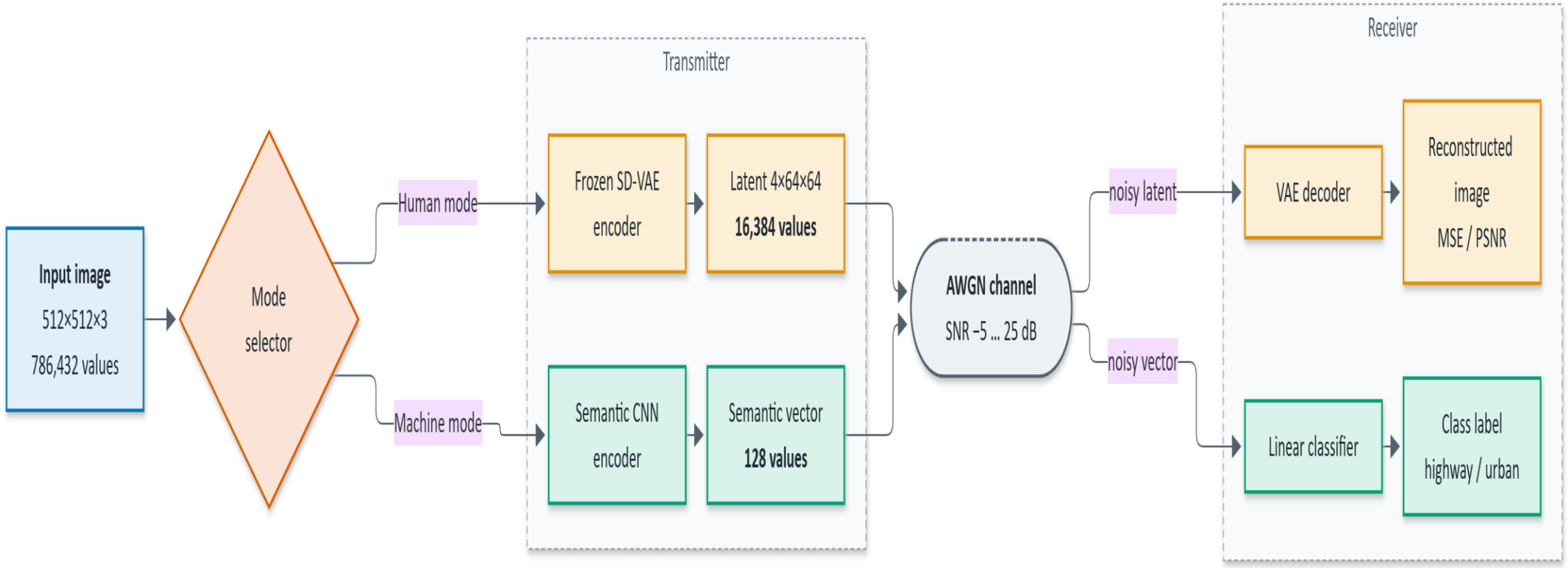
Project Objectives

- Design a dual-branch semantic transmitter: human mode (VAE latent) and machine mode (128-d semantic vector).
- Model the wireless link as a power-normalized AWGN channel over SNR −5 to 25 dB.
- Train the machine branch with SNR-randomized training for channel robustness.
- Quantify human-mode fidelity (MSE, PSNR) and machine-mode accuracy across SNR.
- Quantify bandwidth savings against raw transmission.
- Compare JSCC behavior with a conceptual digital outage (cliff) model.

Implementation Details

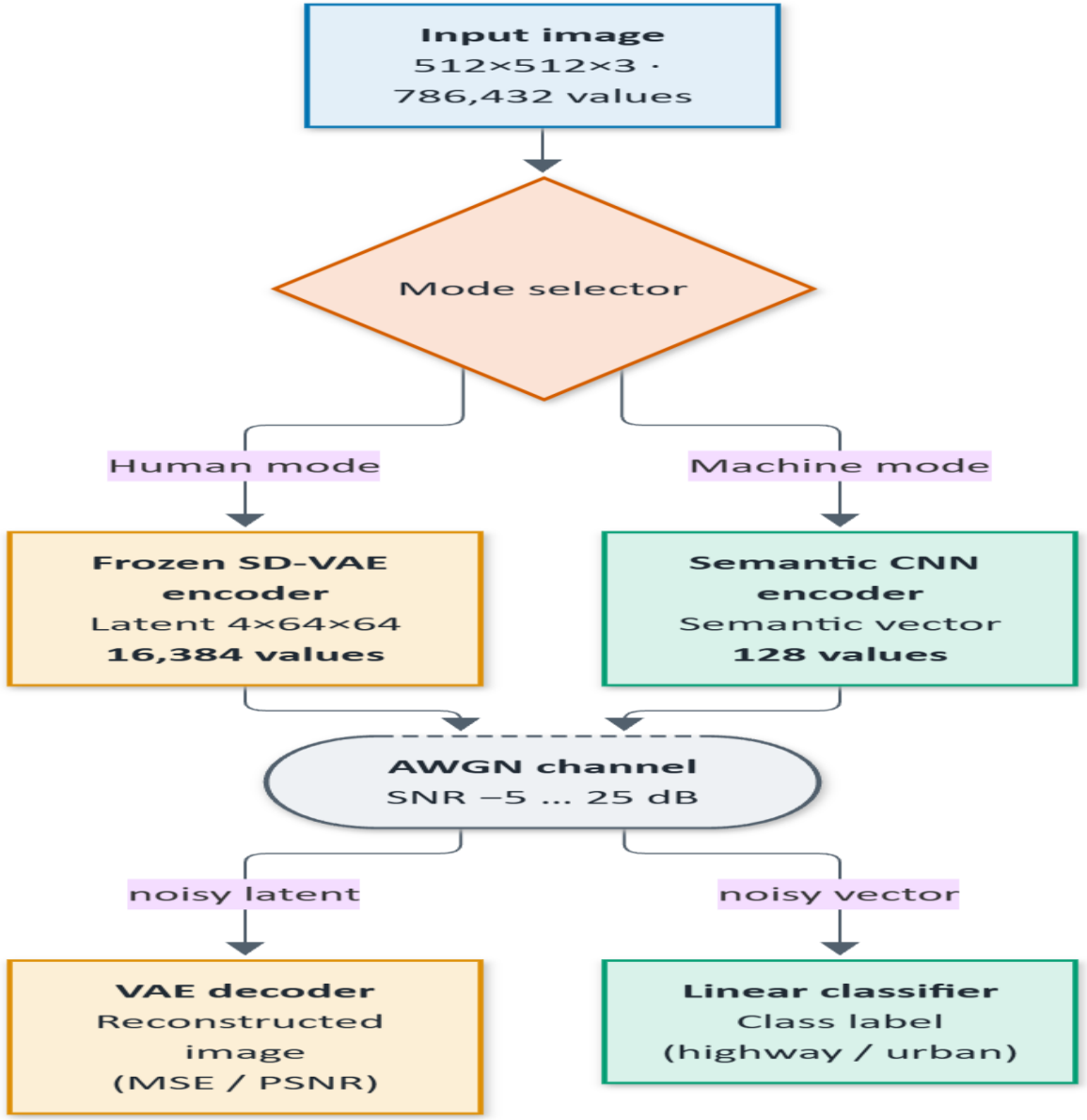
- **Language / framework:** Python 3, PyTorch, torchvision
- **Libraries:** Hugging Face diffusers (AutoencoderKL), scikit-learn, NumPy, Matplotlib, Seaborn
- **Platform:** Jupyter Notebook
- **Pretrained model:** stabilityai/sd-vae-ft-mse
- **Dataset:** traffic_data, 200 images, 2 classes (highway, urban), resized to 512×512
- **Split:** 140 train (87/53), 60 machine test (37/23), 20 human test (12/8)
- **Training:** AdamW, lr 1e-3, weight decay 1e-4, 80 epochs, cross-entropy, seed 6
- **Status:** dual architecture, AWGN channel, training, evaluation and Figures 1–6 complete; digital baseline is conceptual

Block Diagram

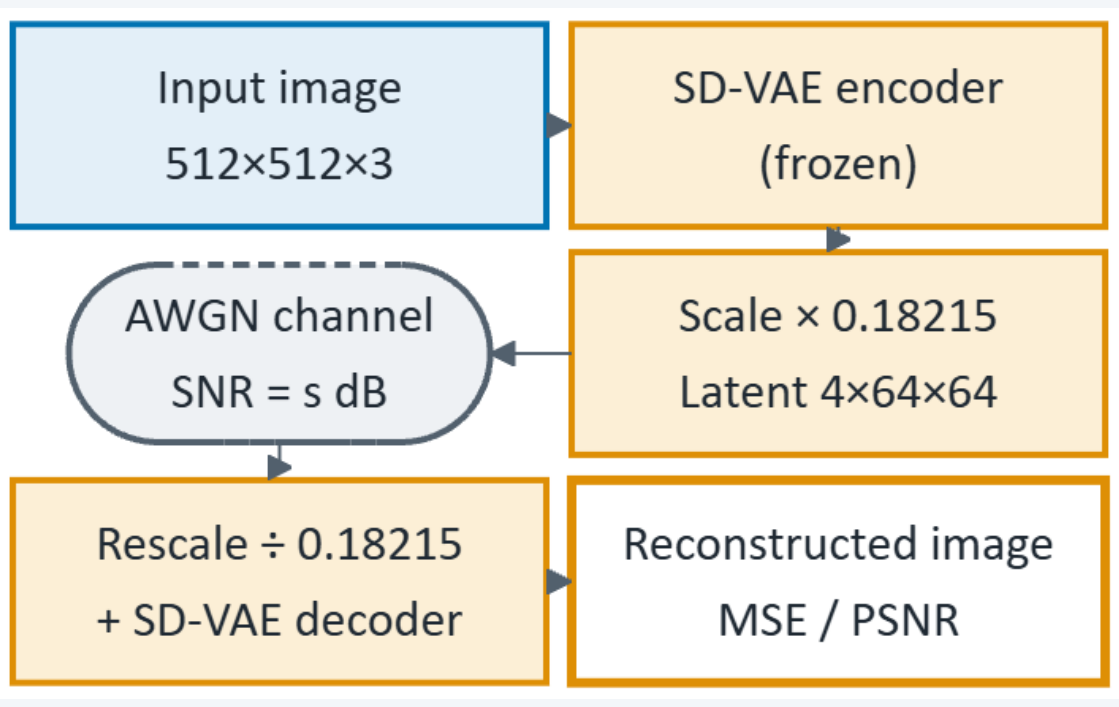


System Overview

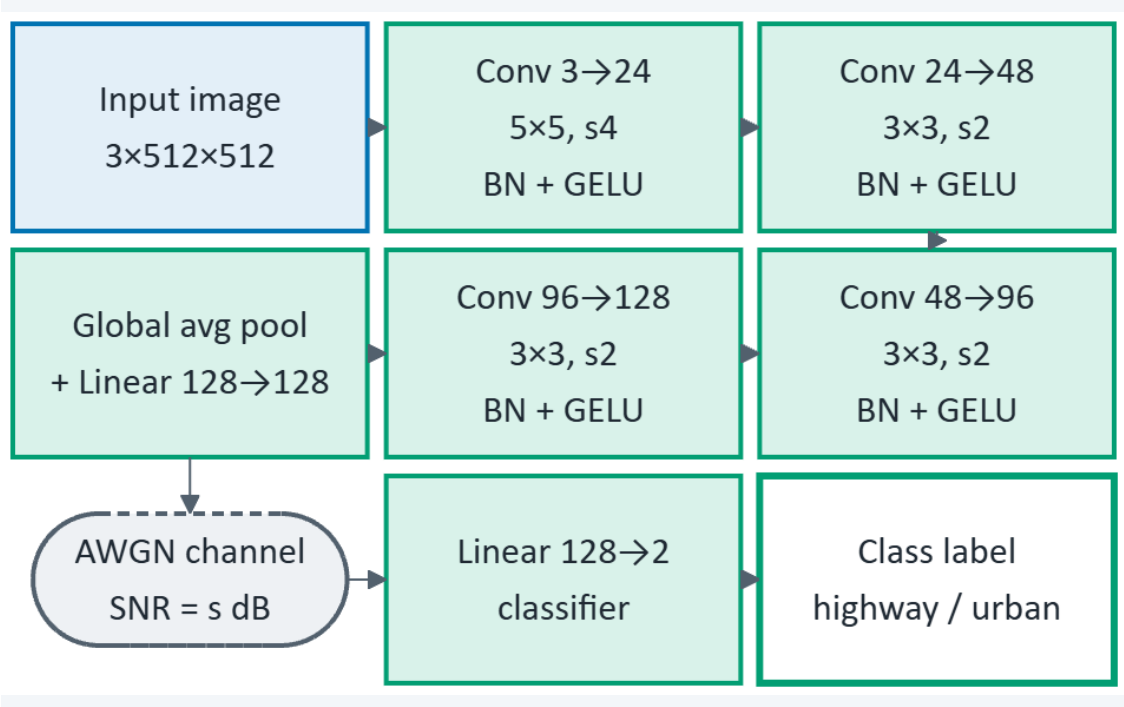
- **Transmitter:** takes a 512×512×3 image and selects the mode.
- **Human mode:** frozen VAE encoder → 4×64×64 latent (16,384 values).
- **Machine mode:** lightweight CNN → 128 values.
- **Channel:** AWGN, noise power = signal power / 10^(SNR/10).



System Overview: Receiver Branches



Human mode: VAE decoder reconstructs the image from the noisy latent.



Machine mode: linear classifier outputs highway / urban from the noisy 128-d vector. Sanity check: empirical SNR measured against target.

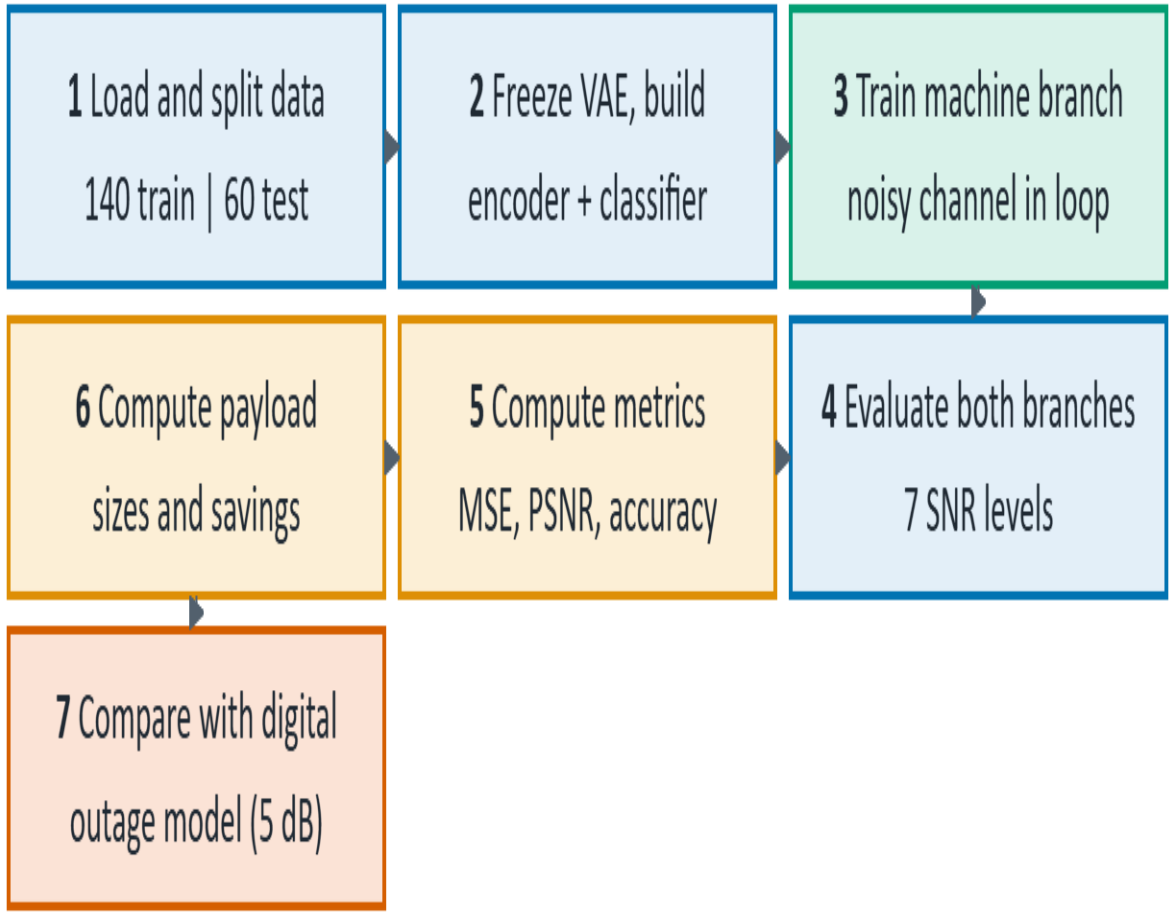
Design: Semantic Encoder and Classifier (Machine Mode)

Layer	Configuration
Input	3 × 512 × 512, normalized to [−1, 1]
Conv1	3 → 24, 5×5, stride 4, BatchNorm, GELU
Conv2	24 → 48, 3×3, stride 2, BatchNorm, GELU
Conv3	48 → 96, 3×3, stride 2, BatchNorm, GELU
Conv4	96 → 128, 3×3, stride 2, BatchNorm, GELU
Pool + Linear	Global average pool → Linear(128 → 128): transmitted vector
Classifier head	Linear(128 → 2)

AWGN-augmented training: one SNR drawn at random per batch from {25, 20, 15, 10, 5, 0, −5} dB.

Methodology

- Load dataset, resize, normalize, stratified split.
- Freeze the VAE; build the semantic encoder and classifier.
- Train the machine branch with the noisy channel in the loop.
- Evaluate both branches at 7 SNR levels on held-out images.
- Compute MSE, PSNR = −10·log10(MSE) and accuracy.
- Compute payload sizes and savings.
- Compare with the conceptual digital outage model (5 dB).



Algorithm

```
Training (machine mode)
for each epoch:
  for each batch (x, y):
    s  <- random SNR from S
    z  <- Encoder(x)
    z' <- z + N(0, P(z)/10^(s/10))
    loss <- CE(Classifier(z'), y)
    update Encoder, Classifier (AdamW)
```

```
Inference
Human:
  l  <- VAE.enc(x) * 0.18215
  l' <- AWGN(l, snr)
  x^ <- VAE.dec(l' / 0.18215)
Machine:
  z  <- Encoder(x)
  y^ <- argmax Classifier(AWGN(z, snr))
```

Novelty

- **Dual-mode framework:** one design serves both human and machine receivers with different payloads.
- **Extreme payload reduction:** 97.92% (human) and 99.98% (machine) fewer values than raw.
- **Channel-robust training:** randomized-SNR AWGN augmentation.
- **Cliff comparison:** explicit comparison of graceful degradation against a digital cliff model.
- **Lightweight and reproducible:** seeded notebook, small CNN, pretrained VAE, no heavy training.

Results 1: Human-Mode Reconstruction

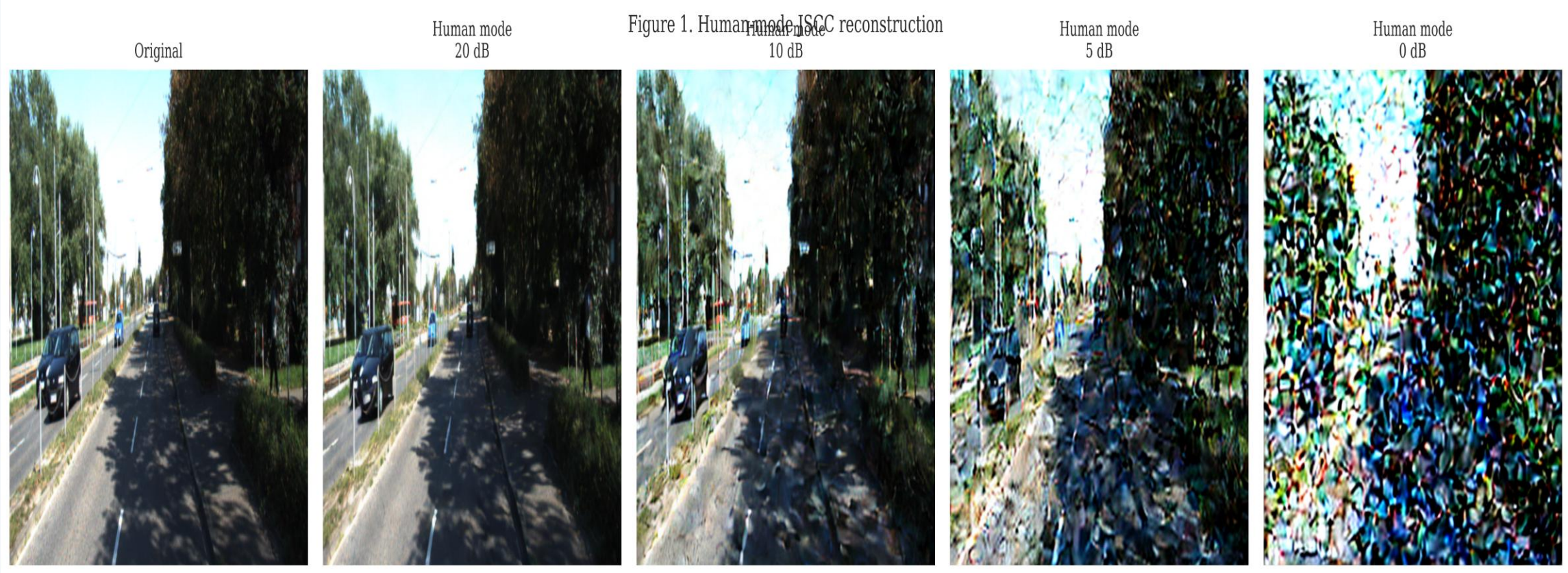
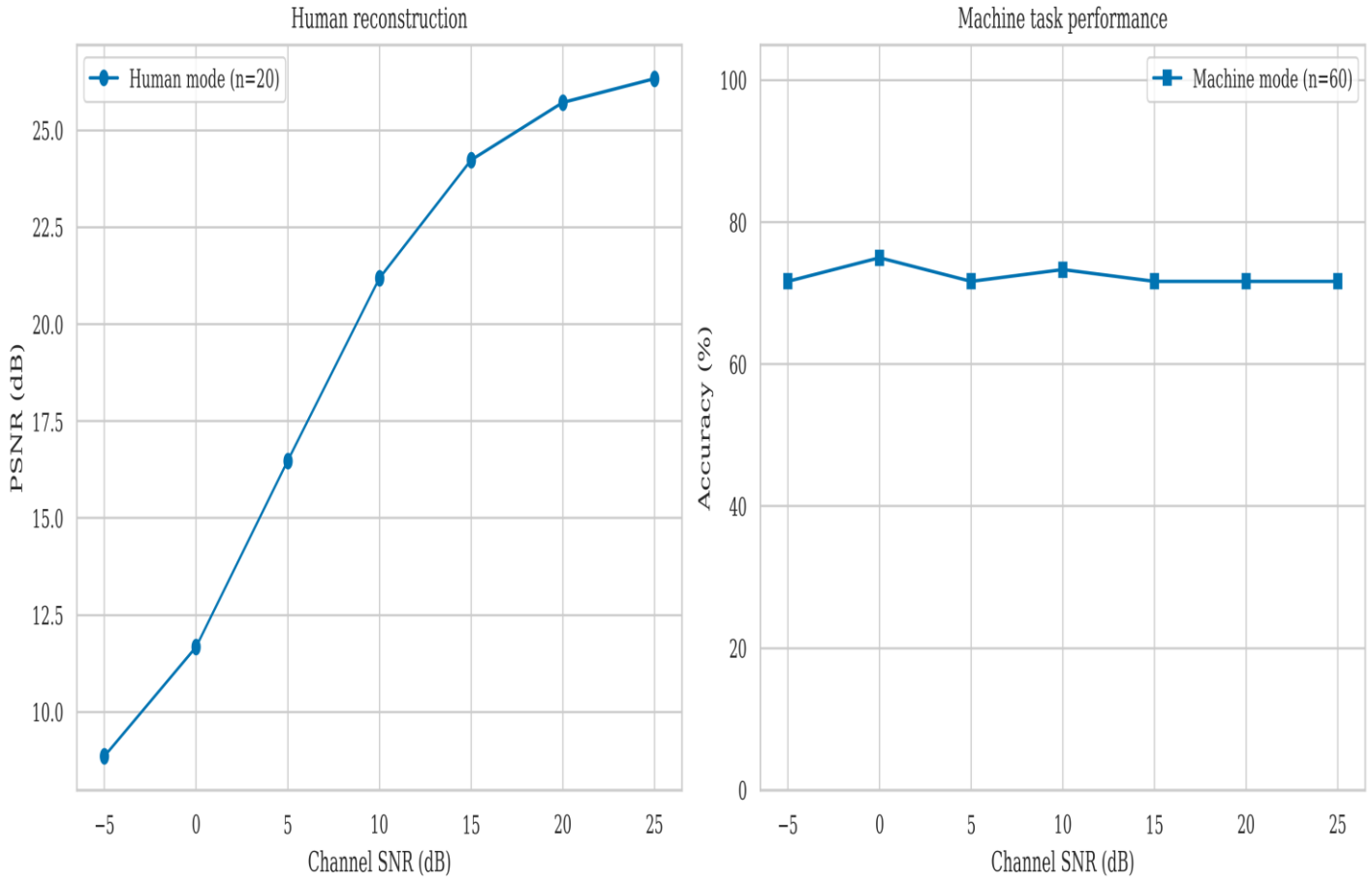


Figure 1. Human-mode ISCC reconstruction

Original and reconstructions at 20, 10, 5 and 0 dB. Quality degrades smoothly as SNR drops.

Results 2: Performance versus SNR



- **Human mode:** PSNR falls smoothly from 26.34 dB (25 dB SNR) to 8.86 dB (−5 dB SNR); no cliff.
- **Machine mode:** accuracy stays at 71.7–75.0% over the whole SNR range (n = 60).
- **Reading:** accuracy does not collapse as noise grows, showing channel robustness of the semantic vector.

Results 3: Training Behavior and Confusion Matrix

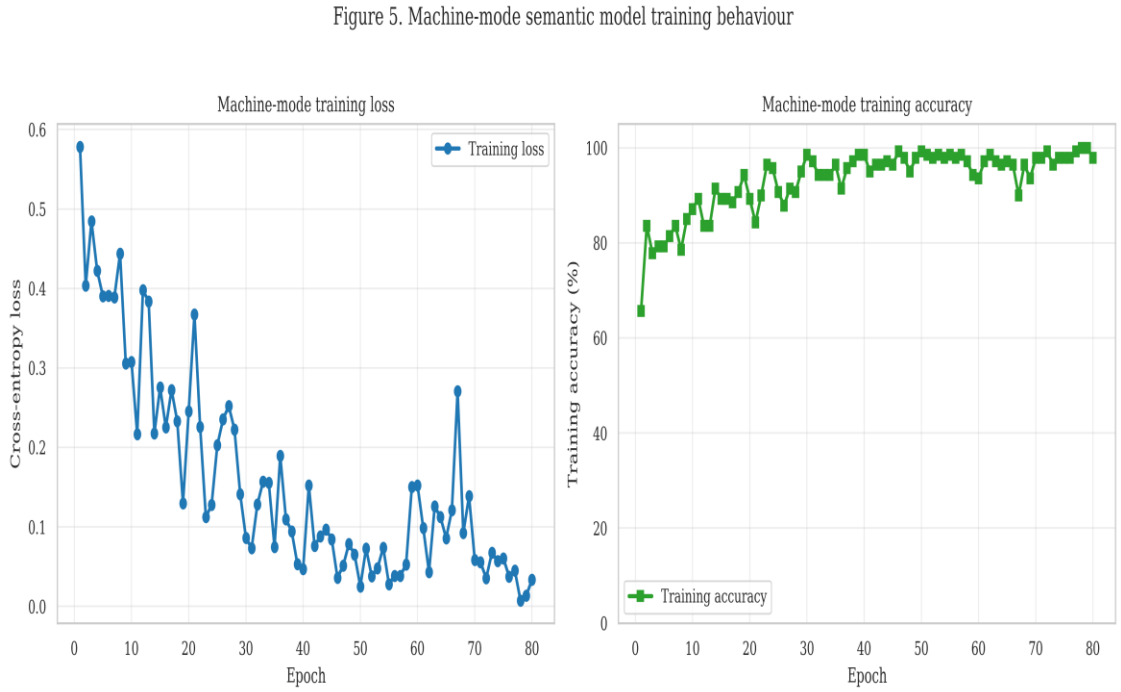


Figure 5. Training loss and accuracy over 80 epochs; training accuracy reaches 98%.

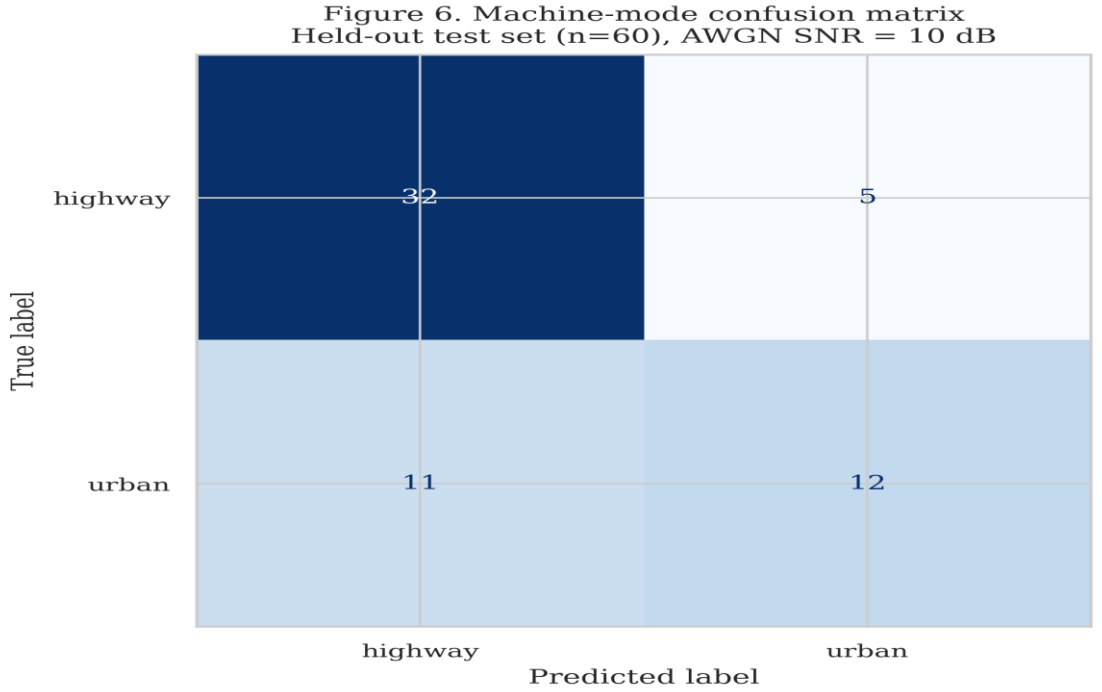


Figure 6. Confusion matrix on the held-out set (n = 60) at 10 dB SNR: 73.33% accuracy. The train–test gap reflects the small dataset.

Results 4: Graceful Degradation vs Digital Cliff

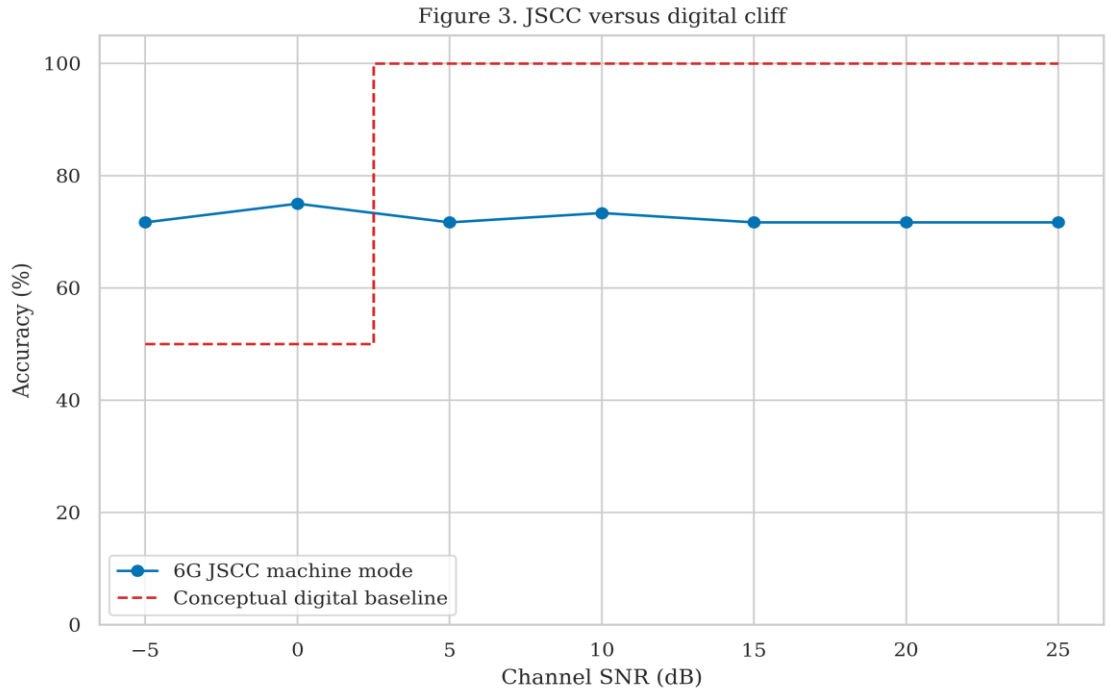
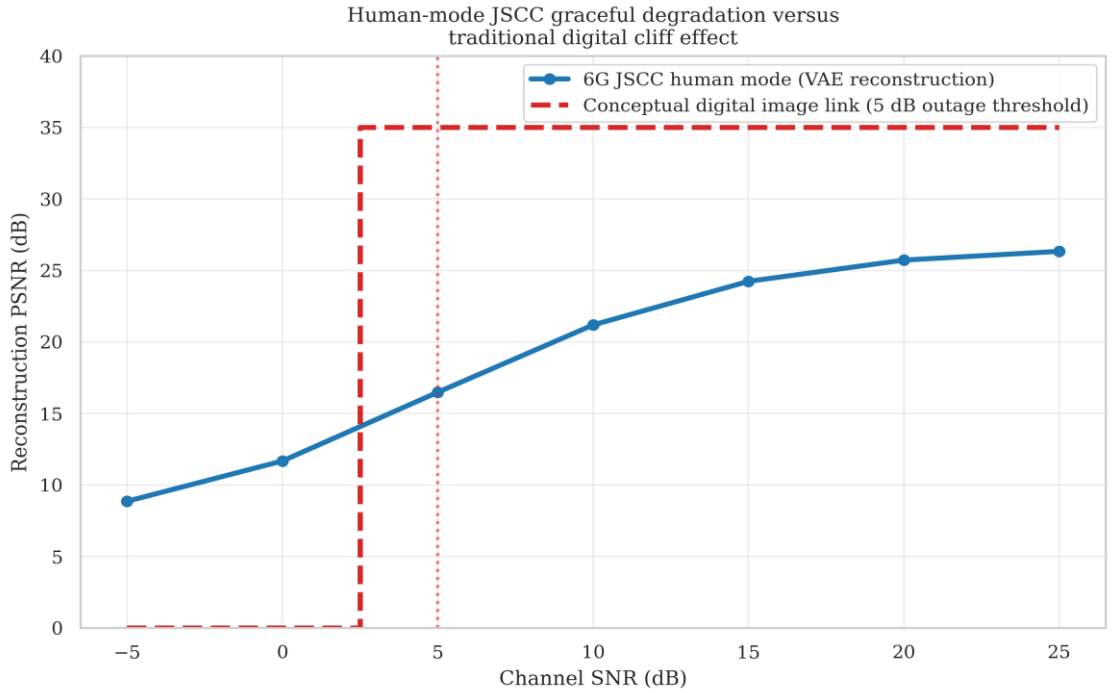
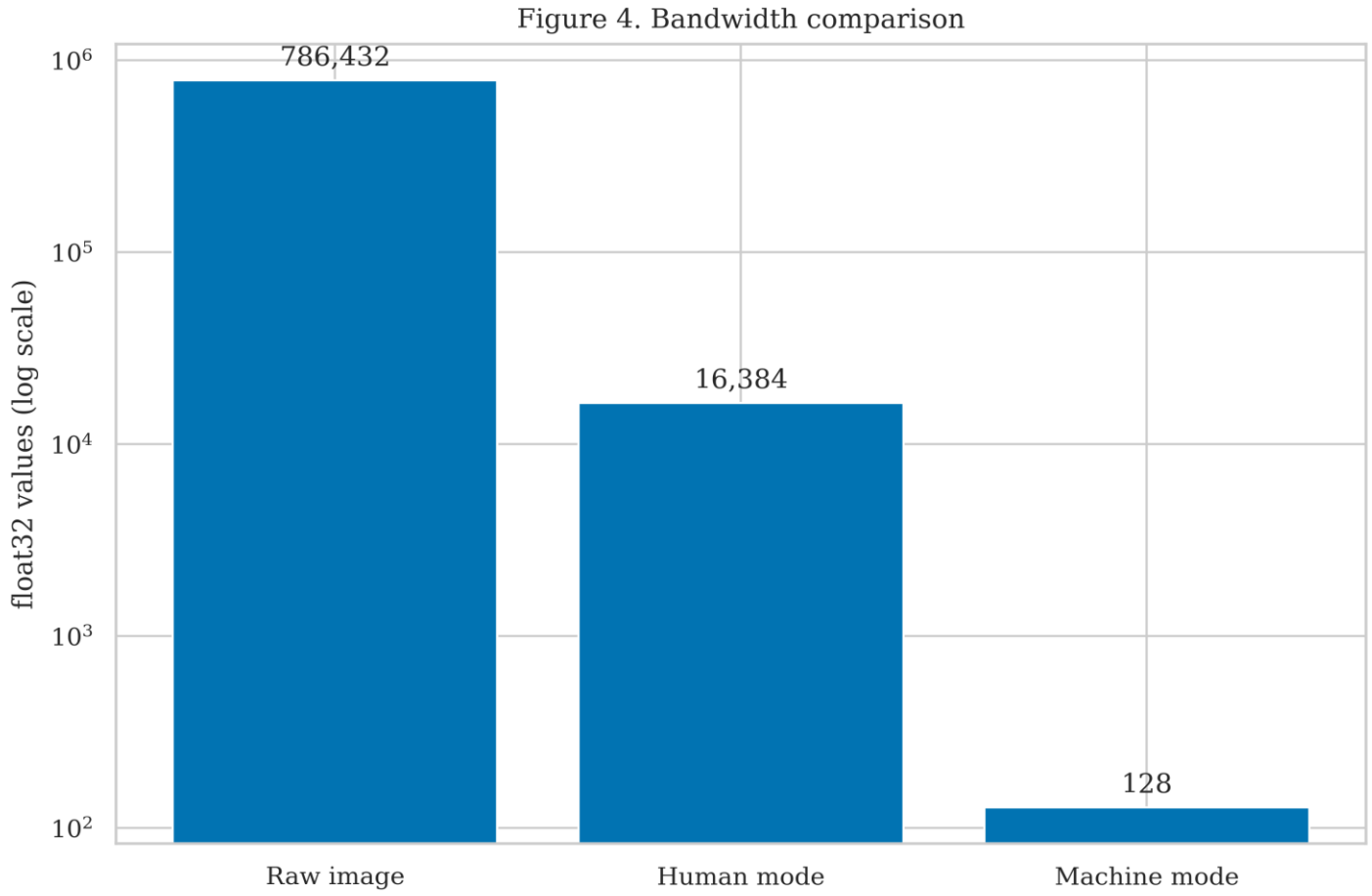


Figure 3. Machine-mode accuracy vs a conceptual digital baseline (assumed 5 dB outage threshold).



Human-mode PSNR vs a conceptual digital image link. JSCC degrades gradually; the digital link does not.

Results 5: Bandwidth Consumption



- **Raw image:** 786,432 float32 values (3.15 MB).
- **Human mode:** 16,384 values (65.5 KB), 97.92% saving, 48× compression.
- **Machine mode:** 128 values (512 B), 99.98% saving, 6,144× compression.
- **Machine vs human:** a further 128× reduction.

SDG Mapping

- **SDG 9: Industry, Innovation and Infrastructure.** Efficient, AI-native 6G communication infrastructure.
- **SDG 11: Sustainable Cities and Communities.** Bandwidth-efficient traffic monitoring for smart cities.
- **SDG 13: Climate Action.** Fewer transmitted values means lower communication energy.
- **SDG 3: Good Health and Well-being.** Faster road-incident detection for safety.

Conclusion and Future Work

- **Conclusion**
- A dual-mode semantic architecture serves human and machine receivers over a noisy 6G-style channel.
- Payload cut by 97.9% (human) and 99.98% (machine).
- Human-mode PSNR degrades smoothly, from 26.34 dB at 25 dB SNR to 8.86 dB at −5 dB.
- Machine accuracy is stable (71.7–75.0%) across SNR.

- **Future work**
- Larger multi-class datasets to close the train–test gap.
- Fading channels and MIMO.
- Trainable human-mode JSCC and a SwinJSCC backbone.
- A simulated digital baseline (JPEG/BPG + LDPC).
- Adaptive rate control and SDR hardware validation.

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Thank You